

The Constants of Nature

$E_{\text{h}}:\text{K}=\text{k}\frac{\left(\frac{T_{\text{p}}}{m_{\text{p}}}\right)}{\frac{m_{\text{p}}}{m_{\text{e}}}}\left(1-14C_{\text{U}}^{-4}\kappa_{\text{p}}\right)\boxtimes$	$m_{\text{de}}=\frac{3s}{\kappa_3\kappa_4}\left(\frac{16}{4\pi}\right)\left(m_{\text{n}}+m_{\text{u}}\right)\left(1+\frac{Im(\omega_1)}{\kappa_2}\left(\frac{L_2}{14}\right)\kappa_3^{-2}\boxtimes\right)$	$E_{\text{p}}=\frac{\text{joule}}{\text{kg}}\left(\frac{L_{\text{p}}^2m_{\text{p}}}{t_{\text{p}}^2}\right)\left(1-\frac{1}{2}Re\left(i^{i''}\right)\kappa_3\kappa_4\boxtimes\right)$	$c=\frac{\left(\frac{L_2}{14}\right)}{1+\sqrt{G_{\text{G1}}}\left(\kappa_1-\kappa_2\right)\boxtimes}$	$F_{90}=\text{farad}\left(1+\frac{C_{\text{d}}}{6}\left(\kappa_1+\kappa_3+\kappa_4\right)\boxtimes\right)$	$K_{\text{f}}=\kappa_1^2\frac{\left(\frac{t_{\text{p}}}{l_{\text{p}}}\frac{q_{\text{p}}}{m_{\text{p}}}\right)}{\left(\frac{l_{\text{p}}}{l_{\text{p}}}\frac{m_{\text{p}}}{m_{\text{p}}}\right)}\left(1-\frac{7}{8}\left(\frac{8}{4\pi}\right)^2\left(\kappa_1^2+\kappa_2^2\right)\boxtimes\right)$	$g_{\text{de}}=\frac{2\pi}{K}\left(1+\frac{16}{\sqrt{4\pi}}\left(\kappa_1^2+\kappa_3^2+\kappa_4^2\right)\boxtimes\right)$
$\text{K}:\text{E}_{\text{h}}=\frac{\text{kelvin}}{\kappa_1^4}\left(\frac{m_{\text{p}}}{t_{\text{p}}m_{\text{e}}}\right)\left(1+14C_{\text{U}}^{-4}\kappa_{\text{p}}\right)\boxtimes$	$m_{\text{u}}=\frac{1}{\kappa_3\kappa_4}\sqrt{\frac{16}{4\pi}}\left(m_{\text{n}}+m_{\text{u}}\right)\left(1+\left(\frac{16}{2\pi}\right)\kappa_3^{-2}\boxtimes\right)$	$R_{\text{K}}=\frac{2\pi}{\kappa_1^2}\left(\frac{L_{\text{p}}^2m_{\text{p}}}{t_{\text{p}}q_{\text{p}}}\right)\left(1+\frac{1}{2}Re\left(i^{i''}\right)\kappa_3\kappa_4\boxtimes\right)$	$\frac{1}{\text{m}}:\text{Hz}=\frac{1}{\text{meter}}\left(\frac{1}{t_{\text{p}}}\right)\left(1+\sqrt{G_{\text{G1}}}\left(\kappa_1-\kappa_2\right)\boxtimes\right)$	$H_{00}=\text{henry}\left(1-\frac{C_{\text{d}}}{6}\left(\kappa_1+\kappa_3+\kappa_4\right)\boxtimes\right)$	$e=\kappa_1\left(\frac{t_{\text{p}}q_{\text{p}}}{l_{\text{p}}^2m_{\text{p}}}\right)\left(1-\frac{7}{8}\left(\frac{8}{4\pi}\right)^2\left(\kappa_1^2+\kappa_2^2\right)\boxtimes\right)$	$\frac{\mu_{\text{de}}}{\mu_{\text{N}}}=\frac{2\pi}{8K}\left(1+\frac{16}{\sqrt{4\pi}}\left(\kappa_1^2+\kappa_3^2+\kappa_4^2\right)\boxtimes\right)$
$\Delta_{\text{vcc}}=\frac{E_1}{2\pi}\left(\frac{t_{\text{p}}}{l_{\text{p}}^2m_{\text{p}}}\right)\left(1-\sqrt{\zeta(2)}\kappa_{\text{p}}^2\boxtimes\right)$	$m_{\text{u}}=\left(\frac{\text{A}}{16}\right)\frac{\text{GeV}}{\kappa_1}\left(\frac{t_{\text{p}}^2}{l_{\text{p}}^2}\right)\left(1-3\left(\frac{\text{A}}{16}\right)\kappa_3^{-3}\boxtimes\right)$	$Z_0=4\pi\left(\frac{l_{\text{p}}^2m_{\text{p}}}{t_{\text{p}}}\right)\left(1+\frac{1}{2}Re\left(i^{i''}\right)\kappa_3\kappa_4\boxtimes\right)$	$m_3=\left(m_{\text{n}}-m_{\text{u}}\right)\left(1+\sqrt{\frac{4}{D_{00}}}\kappa_1\kappa_3\kappa_4\boxtimes\right)$	$\Omega_{00}=\text{ohm}\left(1-\frac{C_{\text{d}}}{6}\left(\kappa_1+\kappa_3+\kappa_4\right)\boxtimes\right)$	$\Phi_0=\frac{\pi}{\kappa_1}\frac{t_{\text{p}}^2m_{\text{p}}}{t_{\text{p}}q_{\text{p}}}\left(1+\frac{7}{8}\left(\frac{8}{4\pi}\right)^2\left(\kappa_1^2+\kappa_2^2\right)\boxtimes\right)$	$\frac{\mu_{\text{de}}}{\mu_{\text{B}}}=\frac{2\pi}{8K}\left(\frac{m_{\text{u}}}{m_{\text{e}}}\right)\left(1+\frac{16}{\sqrt{4\pi}}\left(\kappa_1^2+\kappa_3^2+\kappa_4^2\right)\boxtimes\right)$
$\text{eV}:\text{Hz}=\frac{\text{eV}}{2\pi}\left(\frac{t_{\text{p}}}{l_{\text{p}}^2m_{\text{p}}}\right)\left(1-\sqrt{\zeta(2)}\kappa_{\text{p}}^2\boxtimes\right)$	$m_{\text{he}}=\frac{\sqrt{3}}{\kappa_1^2\kappa_2^2}\left(\frac{\text{GeV}}{2\pi}\right)\left(\frac{t_{\text{p}}^2}{l_{\text{p}}^2}\right)\left(1-\frac{3^{-1}}{\kappa_1\kappa_2}\left(\frac{s}{18}\right)\kappa_3^{-3}\boxtimes\right)$	$Z_{\text{p}}=\left(\frac{l_{\text{p}}^2m_{\text{p}}}{t_{\text{p}}}\right)\left(1+\frac{1}{2}Re\left(i^{i''}\right)\kappa_3\kappa_4\boxtimes\right)$	$\sigma_{\text{p}}=\sqrt{\frac{32}{18}}2\pi\kappa_1^4\left(\frac{l_{\text{p}}^4m_{\text{p}}}{t_{\text{p}}^4m_{\text{e}}}\right)^2\left(1-2\sqrt{6}\sqrt{G_{\text{R1}}}\left(\kappa_1-\kappa_2\right)\boxtimes\right)$	$ST_0=\frac{5}{2}+\log\left(\left(\frac{1}{2\pi}\frac{K\text{A}_{\text{mass}}}{l_{\text{p}}^2T_{\text{p}}m_{\text{p}}}\right)^{3/2}\frac{K\text{E}_{\text{p}}}{T_{\text{p}}P_0}\right)\left(1-\frac{1}{12}\left(\frac{s^2}{\kappa_1\kappa_2}\right)\left(\kappa_1-\kappa_3-\kappa_4\right)\boxtimes\right)$	$\frac{h}{e}=\frac{\text{coulomb}}{\text{coulomb}}\left(\frac{l_{\text{p}}^2m_{\text{p}}}{t_{\text{p}}q_{\text{p}}}\right)\left(1+\frac{7}{8}\left(\frac{8}{4\pi}\right)^2\left(\kappa_1^2+\kappa_2^2\right)\boxtimes\right)$	$b'=\frac{v_{\text{peak}}}{2\pi}\left(\frac{1}{t_{\text{p}}T_{\text{p}}}\right)\left(1+\frac{18}{16}\left(\frac{2\pi}{s}\right)^2\left(\kappa_1^2-\kappa_3^2-\kappa_4^2\right)\boxtimes\right)$
$\text{J}:\text{Hz}=\frac{\text{joule}}{2\pi}\left(\frac{t_{\text{p}}}{l_{\text{p}}^2m_{\text{p}}}\right)\left(1-\sqrt{\zeta(2)}\kappa_{\text{p}}^2\boxtimes\right)$	$G=\text{GeV}^2\left(\frac{t_{\text{p}}^2}{l_{\text{p}}^2m_{\text{p}}}\right)^2\left(1-\left(\frac{4\pi}{18}\right)\kappa_3^{-3}\boxtimes\right)$	$\text{Hz}:\frac{1}{\text{m}}=\frac{1}{\text{second}}\left(\frac{t_{\text{p}}}{l_{\text{p}}}\right)\left(1+\frac{1}{2}Im\left(i^{i''}\right)\kappa_3\kappa_4\boxtimes\right)$	$A_{90}=\text{ampere}\left(1+\sqrt{\frac{9}{J_{0.1}}}\kappa_1\kappa_3\kappa_4\boxtimes\right)$	$ST_1=\frac{5}{2}+\log\left(\left(\frac{1}{2\pi}\frac{K\text{A}_{\text{mass}}}{l_{\text{p}}^2T_{\text{p}}m_{\text{p}}}\right)^{3/2}\frac{K\text{E}_{\text{p}}}{T_{\text{p}}P_0}\right)\left(1-\frac{1}{12}\left(\frac{s^2}{\kappa_1\kappa_2}\right)\left(\kappa_1-\kappa_3-\kappa_4\right)\boxtimes\right)$	$c_1=\left(2\pi\right)^2\left(\frac{l_{\text{p}}^4m_{\text{p}}}{t_{\text{p}}^4}\right)\left(1-\left(\frac{4\pi^2}{32}\right)^2\left(\kappa_1^2+\kappa_2^2\right)\boxtimes\right)$	$\kappa_{\text{B}}=\frac{1}{2\pi}\left(\frac{1}{t_{\text{p}}T_{\text{p}}}\right)\left(1+\frac{18}{16}\left(\frac{2\pi}{s}\right)^2\left(\kappa_1^2-\kappa_3^2-\kappa_4^2\right)\boxtimes\right)$
$\text{Hz}:\text{eV}=\frac{2\pi\text{joule}}{\text{second eV}}\left(\frac{l_{\text{p}}^2m_{\text{p}}}{t_{\text{p}}}\right)\left(1+\sqrt{\zeta(2)}\kappa_{\text{p}}^2\boxtimes\right)$	$G=\left(\frac{l_{\text{p}}^3}{t_{\text{p}}^3m_{\text{p}}}\right)\left(1-\left(\frac{4\pi}{18}\right)\kappa_3^{-3}\boxtimes\right)$	$A_{\text{vel}}=\kappa_1^2\left(\frac{lp}{t_{\text{p}}}\right)\left(1-\frac{1}{2}Im\left(i^{i''}\right)\kappa_3\kappa_4\boxtimes\right)$	$C_{90}=\text{coulomb}\left(1+\sqrt{\frac{9}{J_{0.1}^4}}\kappa_1\kappa_3\kappa_4\boxtimes\right)$	$\sigma=\frac{3}{5}\frac{4\pi}{\left(\Gamma(5)\right)^2}\left(\frac{m_{\text{p}}}{t_{\text{p}}^3T_{\text{p}}^4}\right)\left(1+\sqrt{\frac{15}{8}}\left(\frac{2\pi}{\kappa_1\kappa_2}\right)\left(\kappa_1-\kappa_3-\kappa_4\right)\boxtimes\right)$	$c_{11}=4\pi\left(\frac{l_{\text{p}}^4m_{\text{p}}}{t_{\text{p}}^3}\right)\left(1-\left(\frac{4\pi}{32}\right)^2\left(\kappa_1^2+\kappa_2^2\right)\boxtimes\right)$	$\text{K}:\text{Hz}=\frac{\text{kelvin}}{2\pi}\left(\frac{1}{t_{\text{p}}T_{\text{p}}}\right)\left(1+\frac{18}{16}\left(\frac{2\pi}{s}\right)^2\left(\kappa_1^2-\kappa_3^2-\kappa_4^2\right)\boxtimes\right)$
$h=\frac{\text{joule}}{\text{eV}}\left(\frac{l_{\text{p}}^2m_{\text{p}}}{t_{\text{p}}}\right)\left(1+\sqrt{\zeta(2)}\kappa_{\text{p}}^2\boxtimes\right)$	$m_{\text{e}}=\frac{2V_{\text{le}}}{\text{kilogram$					

$t_p = 5.39125836832313 \times 10^{-48}$ s	Planck time
$t_p = 1.61625918175645 \times 10^{-35}$ m	Planck length
$q_p = 1.87545596713962 \times 10^{-18}$ C	Planck charge
$T_p = 1.4167869850795 \times 10^{32}$ K	Planck temperature
$m_p = 2.17642683817579 \times 10^{-8}$ kg	Planck mass
$\Xi = 0.999999199973626 \times 10^{-7}$	hyperbolic inversion boundary
5	number of dimensions
7	break in scale symmetry
$\kappa_1 = 0.0854245431533304 \dots$	1 st hyperbolic partition constant
$\kappa_2 = 3.66756753485501 \dots$	2 nd hyperbolic partition constant
$\kappa_3 = 1.87649603900417 \dots + 0.46615262615972 \dots i$	3 rd hpc
$\kappa_4 = 1.87649603900417 \dots - 0.46615262615972 \dots i$	4 th hpc
$\kappa_5 = -4.847626244916751 \dots$	hyperbolic radius constant
$\kappa_6 = 2.0134652310924 \dots$	hyperbolic radian constant
$G_{01} = 1.01494160640965 \dots$	Geiesking's constant
$V = 0.02983212819330 \dots$	figure eight knot hyperbolic volume
$K = 0.2915955941772190 \dots$	Catalan's constant
$C_d = 0.291560904030818 \dots$	domingo tiling constant
$D_4 = 1.791622681206959 \dots$	dimer constant
$\omega_1 = 0.764974918528596 \dots + 1.32497062714985 \dots i$	omega_1 constant
$\omega_2 = 1.595790437057192 \dots$	omega_2 constant
$t^{(i)} = -0.438282936727032 \dots + 0.360592471871385 \dots i$	power tower of i
$\phi_i = (\frac{-1}{2})^{1/3}$	imaginary unit
$q_g = 1.15872847301812 \dots$	imaginary golden ratio
$g_g = 3.14159265358979 \dots$	tether length for grazing half the unit circle
$P_{up} = 2.29558714939263 \dots$	Archimedes' constant
$s = 5.24411510858421 \dots$	universal parabolic constant
$L = 2.62057555492219 \dots$	arc length of the unit lemniscate
$L_1 = 1.31102877174605 \dots$	lemniscate constant
$L_2 = 0.599070117367796 \dots$	1 st lemniscate constant
$C_0 = 0.847213084793793 \dots$	2 nd lemniscate constant
$C_{00} = 0.8346268417674073 \dots$	ubiquitous constant
$C_{CF} = 0.6977746579649819 \dots$	Gauss's constant
$r_{C1} = 0.928136044598509 \dots$	continued fraction constant
$R_{FF} = 1.2267410720305 \dots$	Ramanujan's 1 st continued fraction constant
$M = -1.718254863963318 \dots$	Fibonacci factorial constant
$e = 2.71828182845904 \dots$	Euler's number
$\gamma = 0.577215664901532 \dots$	Euler-Mascheroni constant
$A = 1.28242712910062 \dots$	Glaser-Kinkelin constant
$S = 0.828252549678847 \dots$	Sierpinski's constant
$J_{01} = 2.40428555695772 \dots$	1 st root of the Bessel function
$C_{DP} = 1.199678640325157 \dots$	Real fixed point of the hyperbolic cotangent
$C_{DD} = 0.73908514023215660 \dots$	Dottie number
$L_{L1} = 0.652743419349189 \dots$	Laplace limit
$\alpha = 2.620978750959589 \dots$	alpha Feigenbaum constant
$\delta_F = 4.66920160910299 \dots$	delta Feigenbaum constant
$P = 1.32471795724474 \dots$	plastic constant
$S = 3.24697960371714 \dots$	silver constant
$C_0 = 0.6249453657196732 \dots$	QRS constant
$C_{00} = 2.03816937970215 \dots$	associated QRS constant
$\chi_{\text{Hafner-Sarnak-McCurley}}$	Hafner-Sarnak-McCurley constant
$T_0 = 2.73150000000000 \times 10^5$ kelvin	freezing point of water
$P_0 = 1.00000000000000 \times 10^2$ pascal	1 bar
$P_1 = 1.0132507547773 \times 10^5$ pascal	standard atmospheric pressure
$E_0 = 109110229711386 \times 10^{-24}$ joules	energy associated with Δ_{CS}
$f_1 = 5.40000000000000 \times 10^{14}$ Hz	frequency of max light wavelength
$\lambda_{\text{peak}} = 4.965114223174427 \dots$	peak λ of spectral radiance/unit conversion
$\nu_{\text{peak}} = 2.82143937212207 \dots$	peak emission of spectral flux/unit frequency
$\ln(x) = \text{natural logarithm}$	$\ln = \text{lumen} = \text{cd} \cdot \text{sr}$
$\log(x) = \text{gamma function}$	$N = \text{newton} = \text{m kg/s}^2$
$\zeta(x) = \text{Riemann zeta function}$	$\mathcal{N} = \text{noether} = \text{m kg/s}$
$\uparrow n = \text{derangement function}$	$\text{Oh} = \text{ohm} = \text{m}^2 \text{ kg/s}^2 \text{ C}^2$
units	composite units
$m = \text{meter}$	$A = \text{ampere} = \text{C/s}$
$s = \text{second}$	$F = \text{farad} = \text{s}^2 \text{ C}^2/\text{m}^2 \text{ kg}$
$C = \text{coulomb}$	$H = \text{henry} = \text{m}^2 \text{ kg/C}^2$
$K = \text{kelvin}$	$\text{Hz} = \text{hertz} = 1/\text{s}$
$\text{kg} = \text{kilogram}$	$J = \text{joule} = \text{m}^2 \text{ kg/s}^2$
derived units	
$\text{MHz} = \text{megahertz} = 1.00000000000000 \times 10^6$ 1/s	
$\text{fm} = \text{femtometer} = 1.00000000000000 \times 10^{-15}$ m	
$\text{eV} = \text{electron-volt} = 1.00000000000000 \dots$ eV	
$\text{MeV} = \text{mega-electron-volt} = 1.00000000000000 \times 10^6$ eV	
$\text{GeV} = \text{giga-electron-volt} = 1.00000000000000 \times 10^9$ eV	
$r_n = \sqrt{\frac{4\pi}{(\Gamma 5)}}^2$	$r_n = K$
$r_n = \sqrt{\frac{C_0}{4}}$	$r_n = \sqrt{\frac{2\pi}{\log(36)}}$
$r_n = \frac{t_p^2}{t_p^2}$	$E_p = \frac{t_p^2}{t_p^2} m_p$
external transition space	Physics Monastery
$V_{24} = \left(\frac{4\pi}{5!} \right) \left(\frac{4\pi}{15} \right) \left(\frac{4\pi}{35} \right) \left(\frac{4\pi}{18} \right) \left(\frac{4\pi}{32} \right) \left(\frac{4\pi}{8} \right)^4$	Thad Roberts